

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	261498	50.037887
1	[Git, Change, Create]	92826	17.762342
2	[Git, Change, Delete]	58957	11.281477
3	[Git, Change, Rename]	24301	4.650019
4	[File, Git]	16199	3.099694
5	[Git, Commit]	15955	3.053004
6	[Java, ByteCode, Parameter]	9236	1.767317
7	[Java, ByteCode, Member, Method]	8939	1.710486
8	[Java, ByteCode, Bound]	6321	1.209529
9	[Java, Value, ByteCode, Annotation]	5389	1.031190
10	[Java, ByteCode, Bound, ParameterizedType]	4895	0.936663
11	[Xml, Element]	2416	0.462304
12	[Java, ByteCode, Member, Field]	2207	0.422312
13	[Xml, Text]	1622	0.310371
14	[Java, ByteCode, Member, Constructor, Method]	1586	0.303483
15	[Java, ByteCode, Bound, WildcardType]	970	0.185610
16	[Java, ByteCode, Member, Method, Lambda]	851	0.162840
17	[Java, ByteCode, Bound, TypeVariable]	848	0.162266
18	[Type, File, Java, ByteCode, JavaType]	690	0.132032
19	[Type, File, Java, ByteCode, Class]	660	0.126292
20	[Json, Key]	544	0.104095
21	[Value, Json, Scalar]	534	0.102181
22	[Git, Change, Copy]	481	0.092040
23	[Type, File, Java, ByteCode, ResolvedDuplicate...	477	0.091274
24	[Java, ByteCode, GenericDeclaration, Member, M...	466	0.089170
25	[Java, Value, ByteCode, Primitive]	329	0.062954
26	[Type, File, Java, ByteCode, ExternalType]	313	0.059893
27	[Author, Git, Person]	304	0.058171
28	[Committer, Git, Person]	247	0.047264
29	[Maven, Dependency]	195	0.037313

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

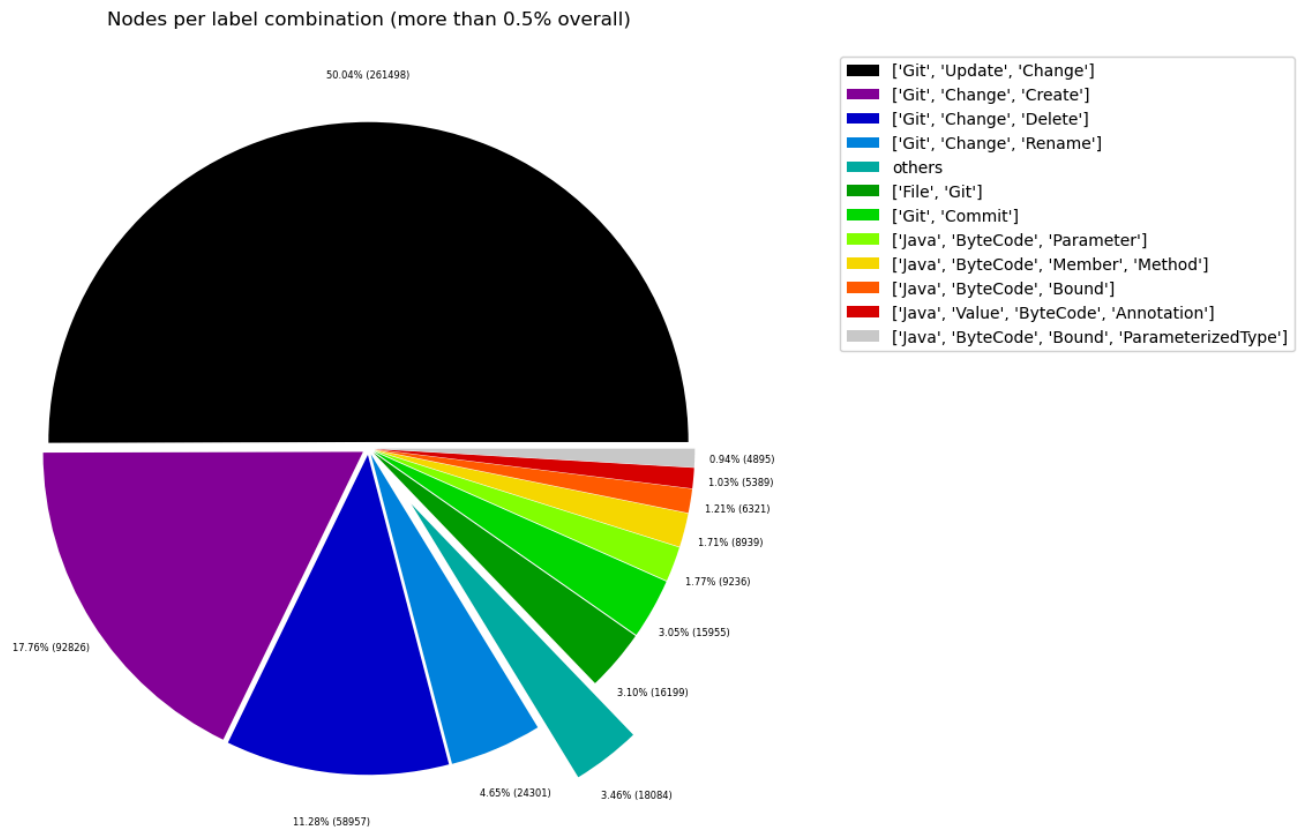


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000191
1	[Git, Branch]	1	0.000191
2	[Repository, File, Git]	1	0.000191
3	[File, Json]	2	0.000383
4	[File]	3	0.000574
5	[Type, File, Java, ByteCode, GenericDeclaratio...	4	0.000765
6	[Maven, Exclusion]	4	0.000765
7	[Value, Array, Json]	8	0.001531
8	[Type, File, Java, ByteCode, Throwable, Extern...	8	0.001531
9	[Java, ByteCode, GenericDeclaration, Member, C...	9	0.001722
10	[Type, File, Java, ByteCode, Throwable, Resolv...	11	0.002105
11	[Maven, Scm]	11	0.002105
12	[File, Maven, Xml, Pom, Document]	11	0.002105
13	[Type, File, Java, ByteCode, Void]	11	0.002105
14	[Java, ManifestSection]	11	0.002105
15	[File, Java, Manifest]	11	0.002105
16	[File, Artifact, Jar, Archive, Zip, Java]	11	0.002105
17	[File, Java, ServiceLoader]	12	0.002296
18	[Maven, PluginExecution]	13	0.002488
19	[File, Java, Properties]	13	0.002488
20	[Maven, ExecutionGoal]	13	0.002488
21	[Type, File, Java, ByteCode, Enum]	17	0.003253
22	[jQAssistant, Rule, Concept]	19	0.003636
23	[Maven, Configuration]	20	0.003827
24	[Maven, Plugin]	20	0.003827
25	[Xml, Attribute]	22	0.004210
26	[Type, File, Java, ByteCode, PrimitiveType]	39	0.007463
27	[Type, File, Java, ByteCode, Annotation]	42	0.008037
28	[Type, File, Java, ByteCode, Class, Throwable]	42	0.008037
29	[Xml, Namespace]	44	0.008419

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

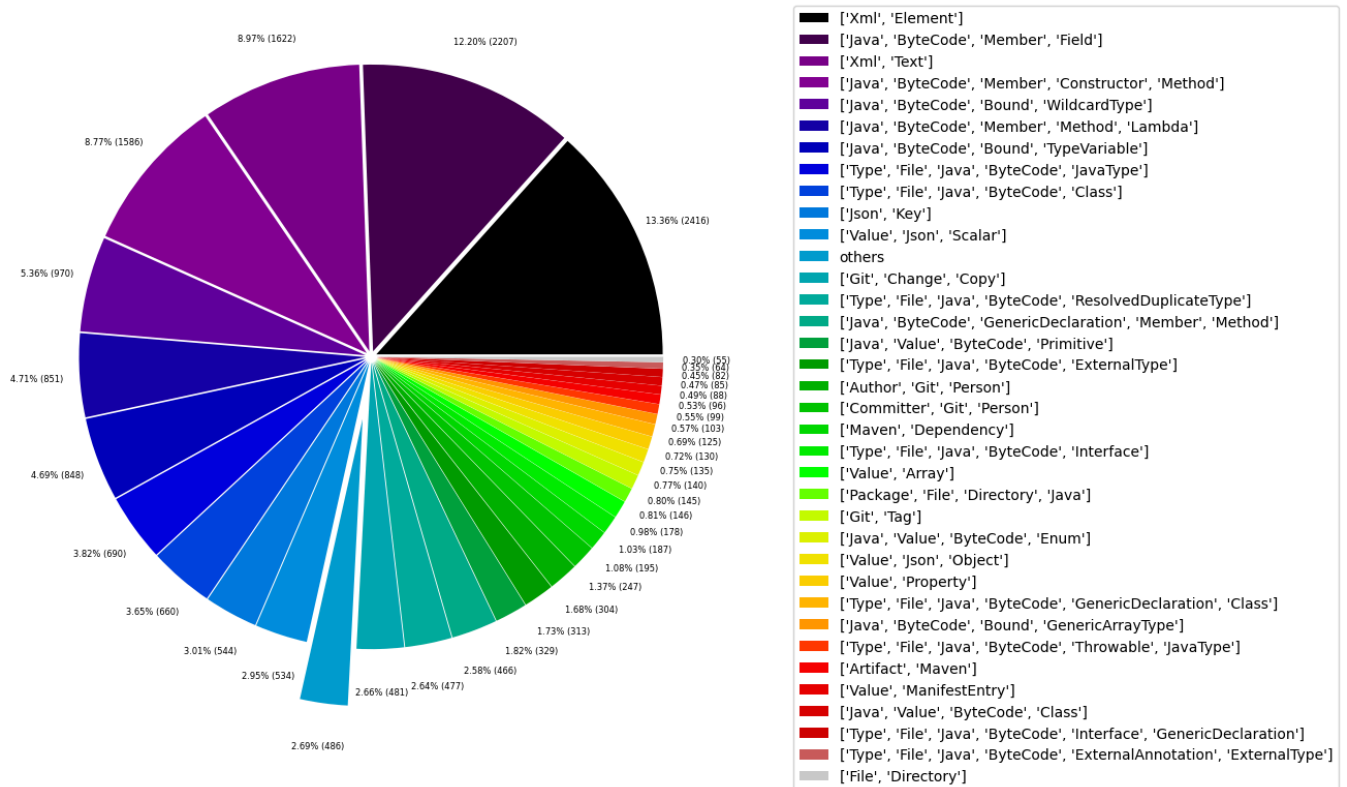


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

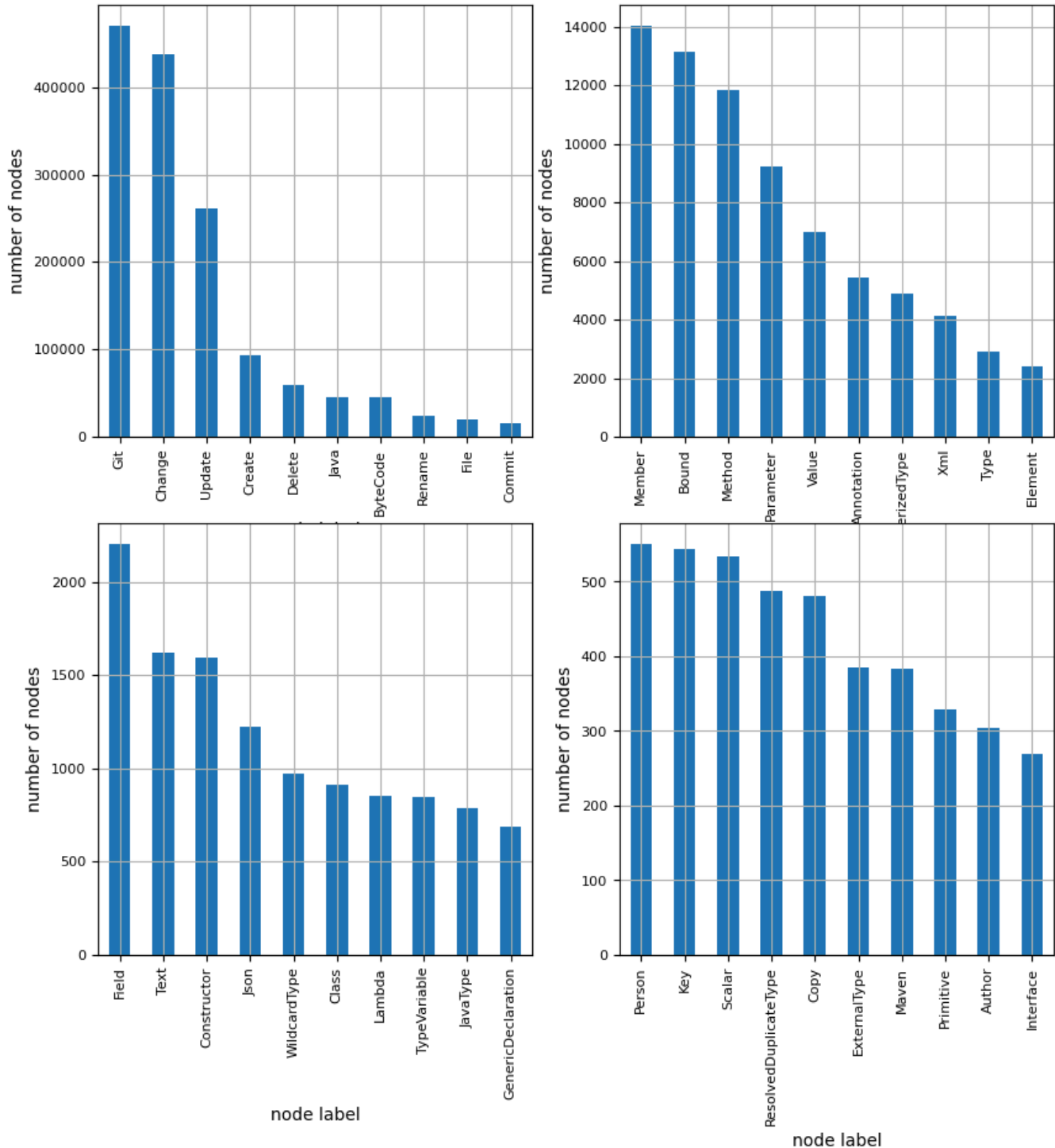
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	470915	90.110027
1	Change	438063	83.823766
2	Update	261498	50.037887
3	Create	92826	17.762342
4	Delete	58957	11.281477
5	Java	45501	8.706659
6	ByteCode	45297	8.667623
7	Rename	24301	4.650019
8	File	19387	3.709721
9	Commit	15955	3.053004
10	Member	14058	2.690011
11	Bound	13137	2.513777
12	Method	11851	2.267700
13	Parameter	9236	1.767317
14	Value	7016	1.342518
15	Annotation	5431	1.039227
16	ParameterizedType	4895	0.936663
17	Xml	4115	0.787409
18	Type	2923	0.559319
19	Element	2416	0.462304
20	Field	2207	0.422312
21	Text	1622	0.310371
22	Constructor	1595	0.305205
23	Json	1223	0.234022
24	WildcardType	970	0.185610
25	Class	912	0.174512
26	Lambda	851	0.162840
27	TypeVariable	848	0.162266
28	JavaType	789	0.150976
29	GenericDeclaration	686	0.131267
30	Person	551	0.105434
31	Key	544	0.104095
32	Scalar	534	0.102181
33	ResolvedDuplicateType	488	0.093379
34	Copy	481	0.092040
35	ExternalType	385	0.073670
36	Maven	383	0.073287
37	Primitive	329	0.062954
38	Author	304	0.058171
39	Interface	269	0.051473

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1621756

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	438063	27.011647
1	MODIFIES	438063	27.011647
2	UPDATES	261498	16.124374
3	CREATES	117608	7.251892
4	DELETES	83258	5.133818
5	COMMITTED	31910	1.967620
6	RENAMES	24301	1.498437
7	INVOKES	22468	1.385412
8	HAS_PARENT	19403	1.196419
9	OF_TYPE	18564	1.144685
10	DEPENDS_ON	17816	1.098562
11	HAS_FILE	16199	0.998856
12	HAS_COMMIT	15955	0.983810
13	DECLARES	14406	0.888296
14	OF_RAW_TYPE	13189	0.813254
15	HAS	9866	0.608353
16	HAS_NEW_NAME	9446	0.582455
17	RETURNS	8167	0.503590
18	READS	6080	0.374902
19	HAS_ACTUAL_TYPE_ARGUMENT	5811	0.358315
20	ANNOTATED_BY	5388	0.332232
21	OF_GENERIC_TYPE	4376	0.269831
22	CHANGED_TOGETHER_WITH	4036	0.248866
23	SIMILAR	3419	0.210821
24	CONTAINS	3227	0.198982
25	RESOLVES_TO	2627	0.161985
26	RETURNS_GENERIC	2601	0.160382
27	HAS_FIRST_CHILD	2416	0.148974
28	HAS_LAST_CHILD	2416	0.148974
29	HAS_ELEMENT	2394	0.147618

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

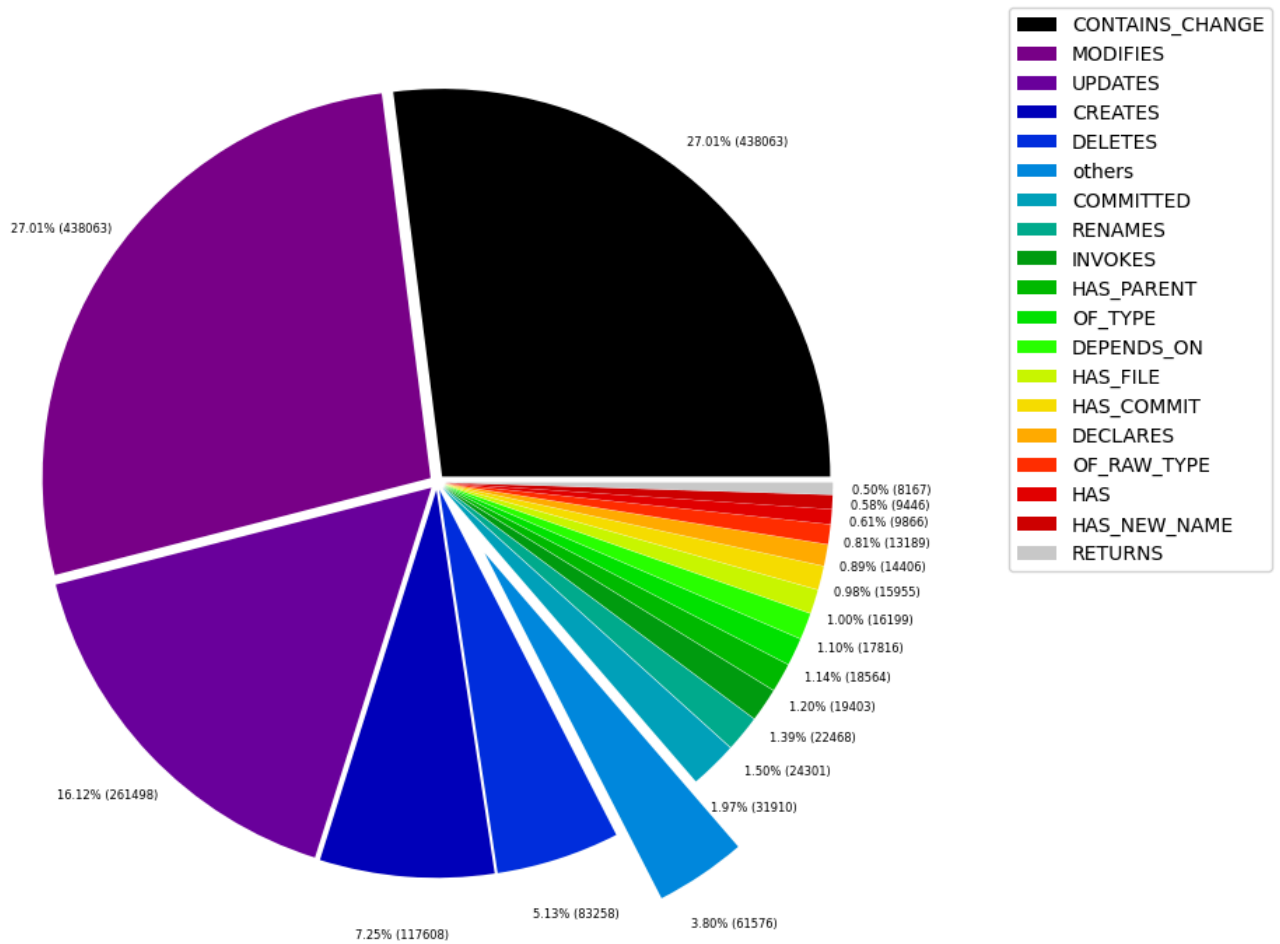


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

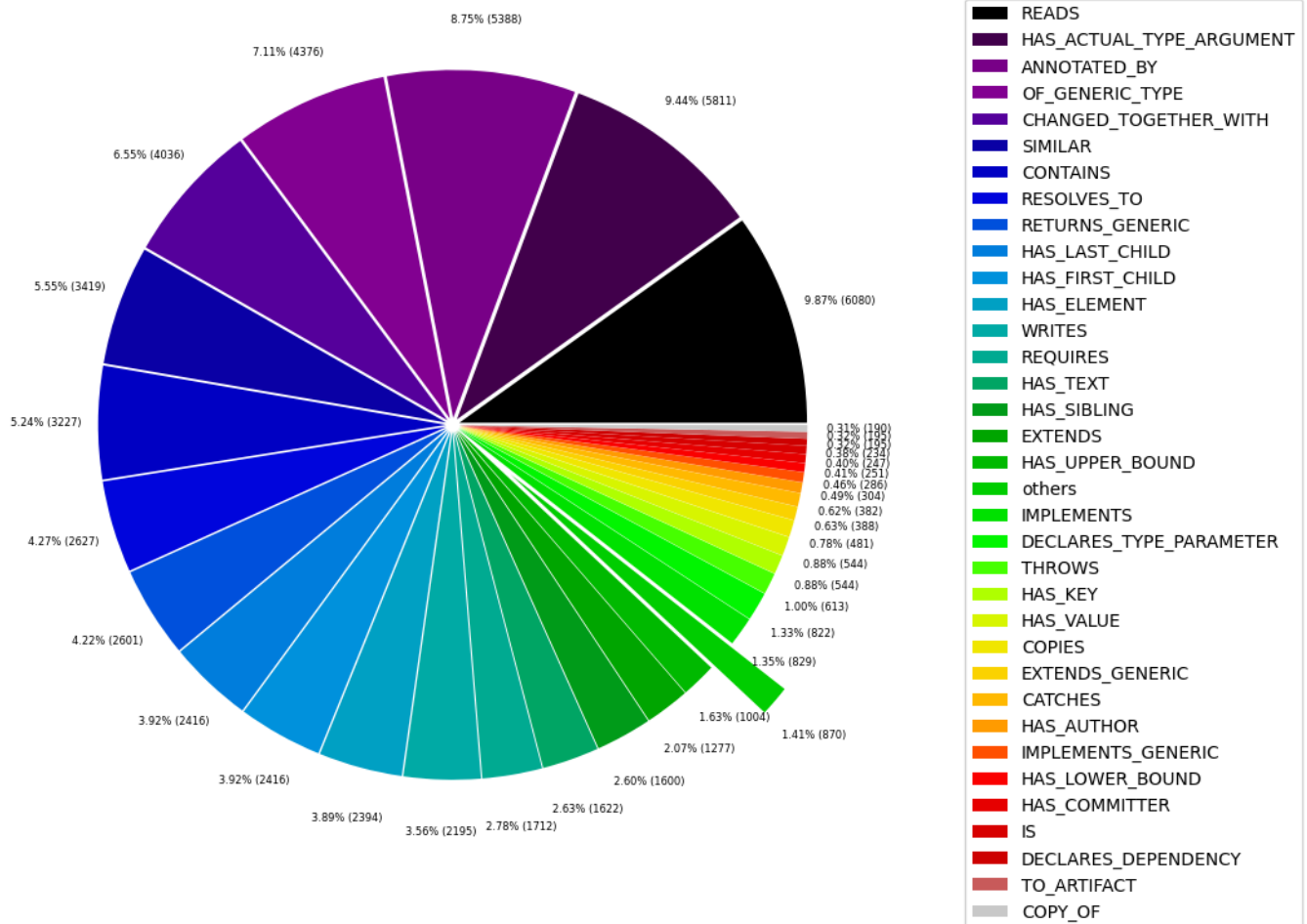
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000062
1	HAS_BRANCH	1	0.000062
2	HAS_HEAD	2	0.000123
3	EXCLUDES	4	0.000247
4	THROWS_GENERIC	5	0.000308
5	HAS_SCM	11	0.000678
6	HAS_ROOT_ELEMENT	11	0.000678
7	DESCRIBES	11	0.000678
8	HAS_GOAL	13	0.000802
9	HAS_EXECUTION	13	0.000802
10	INCLUDES_CONCEPT	19	0.001172
11	IS_ARTIFACT	20	0.001233
12	HAS_CONFIGURATION	20	0.001233
13	USES_PLUGIN	20	0.001233
14	OF_NAMESPACE	22	0.001357
15	HAS_ATTRIBUTE	22	0.001357
16	REQUIRES_TYPE_PARAMETER	26	0.001603
17	REQUIRES_CONCEPT	28	0.001727
18	DECLARES_NAMESPACE	44	0.002713
19	HAS_DEFAULT	53	0.003268
20	HAS_COMPONENT_TYPE	103	0.006351
21	CONTAINS_VALUE	131	0.008078
22	HAS_TAG	145	0.008941
23	ON_COMMIT	145	0.008941
24	COPY_OF	190	0.011716
25	TO_ARTIFACT	195	0.012024
26	DECLARES_DEPENDENCY	195	0.012024
27	IS	234	0.014429
28	HAS_COMMITTER	247	0.015230
29	HAS_LOWER_BOUND	251	0.015477

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberO
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	261498		15955
1	[Git, Update, Change]	UPDATES	[File, Git]	261498		261498
2	[Git, Update, Change]	MODIFIES	[File, Git]	261498		261498
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	92826		15955
4	[Git, Change, Create]	MODIFIES	[File, Git]	92826		92826
5	[Git, Change, Create]	CREATES	[File, Git]	92826		92826
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	58957		15955
7	[Git, Change, Delete]	MODIFIES	[File, Git]	58957		58957
8	[Git, Change, Delete]	DELETES	[File, Git]	58957		58957
9	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	24301		15955
10	[Git, Change, Rename]	CREATES	[File, Git]	24301		24301
11	[Git, Change, Rename]	RENAMES	[File, Git]	24301		24301
12	[Git, Change, Rename]	DELETES	[File, Git]	24301		24301
13	[Git, Change, Rename]	MODIFIES	[File, Git]	24301		24301
14	[Git, Commit]	HAS_PARENT	[Git, Commit]	19392		15955
15	[Repository, File, Git]	HAS_FILE	[File, Git]	16199		1
16	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	15955		1
17	[Author, Git, Person]	COMMITTED	[Git, Commit]	15955		304
18	[Committer, Git, Person]	COMMITTED	[Git, Commit]	15955		247
19	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	13389		8909
20	[File, Git]	HAS_NEW_NAME	[File, Git]	9446		16199
21	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	5458		8909
22	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	5321		8909
23	[Java, Value, ByteCode, Annotation]	OF_TYPE	[Type, File, Java, ByteCode, ExternalAnnotatio...	5035		5389
24	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, javaType]	4123		9236
25	[Java, ByteCode, Parameter]	ANNOTATED_BY	[Java, Value, ByteCode, Annotation]	3821		9236
26	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	3341		16199
27	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	2826		4895
28	[Xml, Element]	HAS_ELEMENT	[Xml, Element]	2394		2416
29	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	2297		6321

Graph Density

```
total_number_of_nodes (vertices): 522600
total_number_of_relationships (edges): 1621756
-> total directed graph density: 5.938100363571358e-06
-> total directed graph density in percent: 0.0005938100363571358
```